```
#Step 1: Preloaded Feedbacks (Given Data)
feedback_data = {
'S_No': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
'Name': ['Ravi', 'Meera', 'Sam', 'Anu', 'Raj', 'Divya', 'Arjun', 'Kiran', 'Leela', 'Nisha'],
'Feedback': [
'  Very GOOD Service!!!',
'poor support,   not happy   ',
'GREAT experience! will come again.',
'okay     okay...',
'  not   BAD',
'Excellent care, excellent staff!',
'good food and good ambience!',
'Poor response and poor handling of issue',
'Satisfied. But could be better.',
'Good support... quick service.'
],
'Rating': [5, 2, 5, 3, 2, 5, 4, 1, 3, 4]
}
print(feedback_data)
```

```
{'S_No': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10], 'Name': ['Ravi', 'Meera', 'Sam', 'Anu', 'Raj', 'Divya', 'Arjun', 'Kiran', 'Leela',
```

```
#Step 2: Add More Feedbacks
#● Ask the user to enter how many more feedbacks they want to add.
#● For each feedback, collect the following inputs from the user:
#o Name
#o Written Feedback (text)
#o Rating (1–5)
# Automatically increment S_No starting from 11 onward.
# Append all new data into the feedback_data dictionary.

N=int(input("how many feedbacks do you want add:" ))
for i in range(N):
  sno=len(feedback_data['S_No'])+1

  Name=input("Enter the name")
  Feedback=input("Enter the feedback")
  Rating=input("Enter the rating(1-5)")
```

```
how many feedbacks do you want add:1
Enter the nameYASH
Enter the feedbackGOOD
Enter the rating(1-5)4
```

```
  feedback_data['S_No'].append(sno)
  feedback_data['Name'].append(Name)
  feedback_data['Feedback'].append(Feedback)
  feedback_data['Rating'].append(Rating)

print(feedback_data)
```

```
{'S_No': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11], 'Name': ['Ravi', 'Meera', 'Sam', 'Anu', 'Raj', 'Divya', 'Arjun', 'Kiran', 'Lee
```

```
#Step 3: Text Cleaning
#Clean all feedback entries as follows:
#● Remove punctuation (.,  ,,  !,  ?)
#● Replace multiple spaces with a single space
#● Remove leading and trailing spaces
#● Convert all text to lowercase

#Tip: Use .replace(), .split() and ' '.join() creatively.

for i in range(len(feedback_data['Feedback'])):
    text = feedback_data['Feedback'][i]

    # Remove punctuation
    text = text.replace('.', '')
    text = text.replace(',', '')
    text = text.replace('!', '')
    text = text.replace('?', '')

    # Remove extra spaces and leading/trailing spaces
    text = ' '.join(text.split())
```

```
    # Convert to lowercase
    text = text.lower()

    # Store cleaned feedback
    feedback_data['Feedback'][i] = text

print("Cleaned Feedback Data:")
print(feedback_data['Feedback'])
```

```
Cleaned Feedback Data:
['very good service', 'poor support not happy', 'great experience will come again', 'okay okay', 'not bad', 'excellent care
```

Double-click (or enter) to edit

```
# Step 4: Word Count Insights

def count_word_in_feedbacks(word):
    count = 0

    for feedback in feedback_data['Feedback']:
        words = feedback.lower().split()

        if word.lower() in words:
            count += 1

    return count


# Print word count insights
print("Feedbacks containing 'good':",
      count_word_in_feedbacks("good"))

print("Feedbacks containing 'poor':",
      count_word_in_feedbacks("poor"))

print("Feedbacks containing 'excellent':",
      count_word_in_feedbacks("excellent"))
```

```
Feedbacks containing 'good': 4
Feedbacks containing 'poor': 2
Feedbacks containing 'excellent': 1
```

```
# Step 5: Final Summary & Insights

# 1. Display final cleaned feedback_data
print("\n--- Final Cleaned Feedback Data ---")
print(feedback_data)

# Convert all ratings to integers
feedback_data['Rating'] = [
    int(rating) for rating in feedback_data['Rating']
]

# Calculate average rating
total_rating = sum(feedback_data['Rating'])
total_feedbacks = len(feedback_data['Rating'])

average_rating = total_rating / total_feedbacks

print("Average Rating:", round(average_rating, 2))



# 3. Find feedback with the longest comment
longest_feedback = max(
    feedback_data['Feedback'],
    key=lambda text: len(text.split())
)

print("\n--- Longest Feedback ---")
print("Comment:", longest_feedback)
print("Word Count:", len(longest_feedback.split()))


# 4. Find unique words across all feedbacks
unique_words = set()
```

```
for feedback in feedback_data['Feedback']:
    words = feedback.split()
    unique_words.update(words)

print("\n--- Unique Words ---")
print(sorted(unique_words))


# 5. Optional: Sort feedbacks by rating (highest to lowest)
print("\n--- Feedbacks Sorted by Rating ---")

sorted_feedbacks = sorted(
    zip(
        feedback_data['S_No'],
        feedback_data['Name'],
        feedback_data['Feedback'],
        feedback_data['Rating']
    ),
    key=lambda item: item[3],
    reverse=True
)

for entry in sorted_feedbacks:
    print(entry)
```

```
--- Final Cleaned Feedback Data ---
{'S_No': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11], 'Name': ['Ravi', 'Meera', 'Sam', 'Anu', 'Raj', 'Divya', 'Arjun', 'Kiran', 'Lee
Average Rating: 3.45

--- Longest Feedback ---
Comment: poor response and poor handling of issue
Word Count: 7

--- Unique Words ---
['again', 'ambience', 'and', 'bad', 'be', 'better', 'but', 'care', 'come', 'could', 'excellent', 'experience', 'food', 'good

--- Feedbacks Sorted by Rating ---
(1, 'Ravi', 'very good service', 5)
(3, 'Sam', 'great experience will come again', 5)
(6, 'Divya', 'excellent care excellent staff', 5)
(7, 'Arjun', 'good food and good ambience', 4)
(10, 'Nisha', 'good support quick service', 4)
(11, 'YASH', 'good', 4)
(4, 'Anu', 'okay okay', 3)
(9, 'Leela', 'satisfied but could be better', 3)
(2, 'Meera', 'poor support not happy', 2)
(5, 'Raj', 'not bad', 2)
(8, 'Kiran', 'poor response and poor handling of issue', 1)
```